Michel Pohl

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Personal Website – Blog
LinkedIn – GitHub
Hugging Face – Google Scholar

Machine learning researcher and engineer

Skills

- **Technical skills:** Python (TensorFlow, PyTorch, Transformers, Optuna, scikit-learn, pandas), C, Matlab, Linux, Git, Docker, AWS.
- **Languages:** French (native), English (fluent), Japanese (JLPT N2).

Professional Experience

Algorithm researcher, **Brainomix** (Oxford, UK) Dec 2022–current

- Developed and deployed a respiratory phase classifier for chest CT using shape features from deep learning-based trachea/lung segmentations, supporting in-house imaging biomarker reliability. Investigated model explainability by examining feature importance, PDPs, and ICE plots.
- Conducted survival analysis using random survival forests and Cox proportional hazards models to assess segmentation quality and ILD biomarker predictive power. Improved ML validation workflows in CI/CD by automating survival analysis and reducing mean runtime by 11%.
- Improved trachea/lung segmentation postprocessing using a level-set-based method, reducing worst-case lung volume estimation error by 26%.
- Explored denoising algorithms such as guided filtering for robust airway segmentation, improving Dice score by 0.055 at specific noise levels.

Visiting researcher, **Sony Computer Science Laboratories** (Tokyo / remote) June–Nov 2022

- Explored NLP methods, e.g., transformer-based models (BERT, MPNet) and LSTMs with GloVe embeddings (in TensorFlow/PyTorch), for fake-news detection, ESG risk-score regression, and analysis of fund-report alignment with ESG criteria to support Nomura Asset Management.
- Designed an interpretable method for fund-level greenwashing detection by deriving projection-distance formulas to compare neural embeddings of internal sustainability reports and external news in a subspace defined using ESG criteria.
- Developed an optical-flow-inspired method to estimate time-varying loadings in security factor models. Trained a VAE to learn and visualize low-dimensional fund-exposure embeddings from per-stock active weights and loadings, supporting GPIF, a major Japanese pension fund.

Computer vision research intern, **Idemia**, *Research and Technology Unit* (Île-de-France, France) Apr–Sept 2016

- Prototyped an end-to-end OCR pipeline in C for degraded text images—segmentation, character recognition, and shortest-path graph decoding using Viterbi's algorithm to select the most likely string—achieving 13.4% higher accuracy than Tesseract 3.04, an open-source OCR engine.
- Designed a method for character over-segmentation (i.e., candidate cut generation) based on Dijkstra's algorithm, modeling the image as a graph with pixels as nodes and edge weights derived from intensities and heuristics. Reduced time complexity via a binary-search-style strategy.
- Implemented a character-level neural-network classifier from scratch (~50k parameters) as part of the OCR pipeline and trained it on ~20k synthetic character images. Improved accuracy by ~5% by incorporating a class-conditional Bayesian aspect-ratio prior.

Physics research intern, **Schlumberger**, *Engineering department* (Kanagawa, Japan) Jan–Feb 2014 and June–Aug 2014

- Modeled fluid flow with PDEs and solved them via harmonic/asymptotic analysis to assess vibration-sensor feasibility for reservoir evaluation.

Education

MS and PhD in Bioengineering, **The University of Tokyo** (Japan) Sept 2016–Sept 2021

- Derived implementations of online training algorithms for RNNs with improved accuracy and complexity; applied them to 3D marker-position forecasting for latency compensation in radiotherapy, achieving competitive results with less training data than prior deep-learning approaches.
- Developed a data-efficient, modular, and interpretable image-forecasting algorithm for MR-guided radiotherapy, combining deformable image registration, PCA-based motion modeling, and temporal dynamics prediction with transformers and online-trained RNNs in PyTorch/Matlab.
- Published 4 articles in Q1 CS journals and 1 article chosen as an Editors' Pick in "Towards Data Science"; the 2 most recent papers extend my PhD work with results from **independent research (2022–2026)** involving new algorithms and experiments (e.g., ablation/statistical analyses).
- Presented my research at 12 conferences in Japan (2 awards), 1 workshop at Seoul National University, and in 1 YouTube video (~450k views).
- Secured competitive scholarships & grants from the Japan Student Services Organization, the Epson International Scholarship Foundation, and UTokyo, e.g., the Doctoral Student Special Incentives Program Type A (top 20 2nd-year PhD applicants at the Graduate School of Engineering).
- Independently completed 9 additional MS-level remote courses in math, alongside UTokyo studies, at Aix-Marseille Université (2017–2019).

MS in Engineering (Diplôme d'ingénieur) and **MS in Signal & Image Processing**, **Centrale Lille / Centrale Méditerranée** (France) 2013–2016

- General engineering and management curriculum followed by a specialization track in AI and computer vision research; valedictorian award.
- Implemented a classifier for graph-structured data combining generalized likelihood-ratio testing with graph Fourier transforms; applied it to Alzheimer's disease diagnosis from PET images by modeling brain regions as graph nodes (3-month research project at Fresnel Institute).
- Built/programmed autonomous robots using C/Arduino (e.g., proportional control) for the French Robotics Cup (2-year project, 6-student team).

BS in Mathematics, Physics, and Computer Science (Classes préparatoires), **Lycée Louis-le-Grand** (Paris, France) 2010–2013

- Highly selective undergraduate science program preparing for nationwide competitive examinations for entry to the best French universities.